

Naive Bayes Foundations: Bayes' Theorem

Contents

1. Overview	2
2. The theorem	2
3. The names of the four parts	2
4. The proof	4
5. Two examples	4
5.1 The dice again	4
5.2 A first look at spam	5
6. Summary	6
7. Sources	6
8. Key terms	6

Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Conditional probability ([Note 82](#)).
- **Leads to:** Naive Bayes ([Note 87](#)).

1. Overview

Key point: Bayes' theorem turns $P(B | A)$ into $P(A | B)$: $P(A | B) = P(B | A) \times P(A) / P(B)$. Naive Bayes is this theorem plus one simplifying assumption.

Bayes' theorem was published in 1763, after Thomas Bayes's death, by his friend Richard Price (Bayes 1763). A whole branch of statistics, **Bayesian statistics**, is built on it (Gelman et al. 2013, §1.3). The theorem is also short and easy to prove.

For machine learning it matters directly: the Naive Bayes classifier is Bayes' theorem combined with the independence assumption of the earlier Note.

2. The theorem

Key point: $P(A | B) = P(B | A) \times P(A) / P(B)$, for $P(B) \neq 0$.

For two events A and B with $P(B) \neq 0$:

$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

The conditional probability Note showed that $P(A | B)$ and $P(B | A)$ are usually different. Bayes' theorem is the rule for getting one from the other.

3. The names of the four parts

Key point: Posterior = likelihood $\times$ prior / evidence.

The diagram illustrates Bayes' theorem with the following components and annotations:

- Posterior:** $P(A | B)$ (red text, red arrow pointing to the expression)
- Likelihood:** $P(B | A)$ (blue text, blue arrow pointing to the expression)
- Prior:** $P(A)$ (green text, green arrow pointing to the expression)
- Evidence:** $P(B)$ (orange text, orange arrow pointing to the expression)

The theorem is expressed as:

$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

Meanings provided in the diagram:

- likelihood:** how probable B is if A is true
- prior:** belief in A before seeing B
- posterior:** belief in A after seeing B
- evidence:** how probable B is overall

Figure 1: The four parts of Bayes' theorem

Part	Name	Meaning
$P(A B)$	posterior	how likely A is after seeing B
$P(B A)$	likelihood	how likely the evidence B is if A is true
$P(A)$	prior	how likely A was before seeing anything
$P(B)$	evidence	how likely the evidence B is overall

The likelihood here is the same idea as in the [log loss Note](#) (section three): the probability of what we observed, given a hypothesis (there, a model's coefficients; here, an event A).

So Bayes' theorem describes how a belief is updated: start from the **prior**, see some evidence, and end with the **posterior**.

An everyday picture: a doctor first guesses how common a disease is among all patients (the prior). A test result arrives (the evidence), and the doctor revises the guess for this one patient (the posterior).

In classification, A will be a class, the **target** (the output we predict, such as "spam"). B will be the observed **features** (the input variables, such as the words in an email). The posterior is what the classifier wants.

4. The proof

Key point: Write $P(A \cap B)$ in two ways using conditional probability and set them equal.

Start from the definition of conditional probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad (1)$$

The same definition with the roles swapped gives

$$P(B | A) = \frac{P(B \cap A)}{P(A)}$$

“Both B and A ” is the same event as “both A and B ”, so $P(B \cap A) = P(A \cap B)$. Multiplying both sides by $P(A)$:

$$P(A \cap B) = P(B | A) \times P(A) \quad (2)$$

Substituting (2) into (1) gives Bayes’ theorem:

$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

5. Two examples

Key point: Bayes recovers $P(\text{die 1} = 5 | \text{sum} \leq 10) = 5/33$ from $P(\text{sum} \leq 10 | \text{die 1} = 5) = 5/6$; and ‘free’ raises the chance of spam from 20% to 75%.

5.1 The dice again

Key point: The theorem gives the same $5/33$ found by counting.

From the conditional probability Note, with A = “die 1 shows 5” and B = “the sum is at most 10”: $P(B | A) = 5/6$, $P(A) = 1/6$ and $P(B) = 11/12$. Then

$$P(A | B) = \frac{\frac{5}{6} \times \frac{1}{6}}{\frac{11}{12}} = \frac{5/36}{33/36} = \frac{5}{33}$$

exactly the value found by counting.

5.2 A first look at spam

Key point: Prior 0.2, likelihoods 0.6 and 0.05: seeing “free” makes spam 75% likely.

Suppose 20% of emails are spam. The word “free” appears in 60% of spam emails and in 5% of normal ones. An email contains “free”: how likely is it to be spam?

- Prior: $P(\text{spam}) = 0.20$.
- Likelihood: $P(\text{free} \mid \text{spam}) = 0.60$.
- Evidence: “free” can appear in spam or in normal email, so $P(\text{free}) = 0.60 \times 0.20 + 0.05 \times 0.80 = 0.12 + 0.04 = 0.16$.

$$P(\text{spam} \mid \text{free}) = \frac{0.60 \times 0.20}{0.16} = 0.75$$

Seeing one word has moved the belief from 20% to 75%. The way the evidence was split into “spam” and “not spam” cases is the subject of the next Note, and combining many words is what Naive Bayes does.

6. Summary

- Bayes' theorem: $P(A | B) = P(B | A)P(A)/P(B)$.
- Posterior = likelihood $\times$ prior / evidence: it updates a prior belief with an observation.
- The proof writes $P(A \cap B)$ two ways using conditional probability.
- Naive Bayes applies the theorem with A = class and B = the observed features.

7. Sources

- **Gelman et al. 2013:** Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A. and Rubin, D. B. *Bayesian Data Analysis*, 3rd ed. CRC Press, 2013. Section 1.3, pp. 6–7.
- **Bayes 1763:** Bayes, T. “An Essay towards Solving a Problem in the Doctrine of Chances.” Communicated by R. Price. *Philosophical Transactions of the Royal Society of London* 53, 370–418, 1763.

8. Key terms

Term	Meaning
Bayes' theorem	$P(A B) = P(B A)P(A)/P(B)$: the rule that reverses a conditional probability
Prior	The probability of an event before any evidence is seen
Likelihood	The probability of the observed evidence if a given event is true
Evidence	The overall probability of the observed evidence
Posterior	The probability of an event after the evidence is taken into account
Bayesian statistics	The branch of statistics that treats probabilities as beliefs updated by Bayes' theorem